JUEN JIE TAY

ROBOTICS | EMBODIED AI | COMPUTER VISION
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PROFILE

Final-year NTU EEE student graduating May 2027, focused on embodied AI and robotics. Experience includes ROS 2 robot development, a UR5e manipulation FYP, and RF deep learning and CPU edge deployment at ST Engineering.

TECHNICAL SKILLS

Languages: Python, C++
ML: PyTorch, LeRobot, Hugging Face Transformers, Claude API, faster-whisper, ONNX, OpenVINO
Robotics: ROS 2, Nav2, SLAM Toolbox, robot_localization (EKF), micro-ROS, MuJoCo, Foxglove, UR5e, RealSense
Signal Processing and Tools: DSP (STFT), USRP SDR, HDF5, Zarr, CMake, Git, Linux, uv, pytest, Sphinx

EXPERIENCE

ST Engineering **Singapore**
Edge AI and Computer Vision Intern **Jan 2026 to Aug 2026**

- Trained and compared CNN, SE-ResNet and Swin-Tiny models for drone detection in RF spectrograms. Applied multiple instance learning to sub-band views and selected SE-ResNet-18 to balance inference speed and accuracy.
- Built a multithreaded C++ engine for USRP capture and OpenVINO CPU inference across the 2.4 GHz and 5.8 GHz bands. Connected capture and inference threads through a bounded queue.
- Developed a C++ PID automatic gain controller with a slew-rate limit and anti-windup that holds the noise floor steady in changing RF conditions, so the model receives a consistent input level; verified with GoogleTest unit tests.
- Created a PySide6 and C++ tool to record, replay, simulate and view SDR captures; moved the training dataset from HDF5 to chunked Zarr and documented the training pipeline.

SELECTED PROJECTS

Modular vs End-to-End VLA Models for Embodied Robotic Manipulation **Aug 2026 to Present**
LeRobot | *SmolVLA* | *OWLv2* | *Claude API* | *RealSense* | *UR5e* | *NTU Final Year Project*

- Designed a UR5e comparison between a modular OWLv2 and LLM pipeline and an end-to-end SmolVLA policy. Both use shared speech input, robot control, camera access and safety checks.
- Implementing command clarification, motion safety checks and failure tracing for the modular pipeline. Integrated SmolVLA through LeRobot, with 230+ unit tests across the project.
- Built an evaluation suite of 5 task tiers (pick-and-place, pouring, ambiguity, changing world state and generalisation) with 20 seeded layouts per tier and held-out phrasings, frozen before tuning either pipeline.

Knowledge Space **Jan 2026 to Present**
Obsidian | *Quartz* | *Cloudflare Pages* | *Personal Project*

- Built a public digital garden of 240 notes across 10 fields with Obsidian, Quartz and Cloudflare Pages.

Autonomous Robotic Tank **Feb 2026 to Jun 2026**
ROS 2 | *C++* | *SLAM* | *Nav2* | *Foxglove* | *Personal Project*

- Built a tracked robot with 2D LiDAR, IMU and encoders; wrote C++ micro-ROS firmware on a Pico W for PID speed control, a command watchdog and an ultrasonic e-stop.
- Fused encoder and IMU data in a 30 Hz EKF; fixed SLAM map drift caused by track slip by moving heading to the IMU, calibrating the wheelbase and zeroing IMU heading at boot.
- Compared DWB and Regulated Pure Pursuit on a Raspberry Pi 4 and selected RPP after DWB missed its control deadlines; the robot replanned around obstacles at 2 Hz with RPP control at 5 Hz.
- Streamed ROS 2 topics to Foxglove over a websocket bridge after Fast DDS and Cyclone DDS discovery over WiFi was unreliable; documented the build and tests on a public Sphinx site.

EDUCATION

Nanyang Technological University, Singapore **Expected May 2027**
Bachelor of Engineering in Electrical and Electronic Engineering | *Data Analysis and Machine Learning*

- **AI and robotics coursework:** Multimodal Large Language Models, Computer Vision, Pattern Recognition and Machine Learning, Robotics and Automation, and Natural Language Processing.
- **Selected results:** Enhanced Professional Internship A+, Modelling and Control A, Signals and Systems A-, Engineering Electromagnetics A-.

ADDITIONAL EXPERIENCE AND LEADERSHIP

Freelance Photographer and Videographer **Oct 2024 to Jan 2026**

- Managed client projects end to end, from sales and requirements to shooting, editing and final delivery.

Certifications Universal Robots e-Series Core Track, Jun 2026 | Google Advanced Data Analytics Specialization, Jul 2024
Leadership Badminton Captain, NTU Hall 10 | Badminton Captain, Yishun Innova Junior College